OFFLINE-FIRST SECURITY ENGINEERING

VNAGY Security Architecture — CRA Alignment Layer (Preparation Draft)

Document ID: VNAGY-CRAAL-010

Version: v1.0

Revision History:

Version	Date	Author	Description
V1.0	03 September 2026	V. Nađ	Draft

Document Status: CRA Alignment Preparation Draft

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1. Introduction & Scope

Purpose of **CRA** Alignment Layer

- **VNAGY** offline-first philosophy
- Non-commercial, independent architecture context
- Document status and limitations
- **CRA** regulatory mapping overview

2. VNAGY System Overview

- Offline-first security engineering
- Local event bus
- Local scoring
- Local cache
- Zero cloud dependency
- Local SIEM exporter
- High-level architecture diagram (textual description)

3. Core Components

3.1. M-Module (Public Mathematical Structure)

Mathematical definition

- Signal ingestion
- Weight-free public model
- Offline-first scoring
- Behavior correlation
- Anti-evasion logic (non-public)

3.2. Behavior Engine

- Five signal sources
- Three-stage detection model
- Memory-only attack detection
- Process-without-file heuristics
- CPU-behavior monitoring module
- Subnetting anomaly heuristics

3.3. VNAGY Killing Chain

- Phase definitions
- MITRE mapping
- KEY tools
- Offline-first correlation pipeline

4. CRA Alignment Layer

4.1. CRA Requirements Overview

Cyber Resilience Act scope

- Security documentation expectations
- Software lifecycle expectations
- Transparency requirements
- Vulnerability handling expectations

4.2. VNAGY → CRA Mapping

Offline-first compliance

- Behavior-based detection compliance
- Supply-chain security compliance
- Logging & retention compliance
- Update & patching model
- Transparency & documentation compliance

5. Operational Risk & Detection Mapping

- Functional risk categories
- Detection coverage
- Behavior-signal mapping
- Operational impact analysis
- CRA-aligned risk table

6. Supply-Chain Security Module

- Hash verification
- Digital signatures
- Dependency freeze
- Offline build pipeline
- USB guard
- Config integrity shield
- Ingestion anti-corruption guard
- CRA compliance mapping

7. VNAGY Rule Format (v1.0)

- Structure
- Syntax
- Offline-first rule evaluation
- CRA transparency alignment
- Example rule (safe, non-sensitive)

8. Logging, Retention & SIEM Export

- Local SQLite retention (7–30 days)
- Offline-first SIEM export
- CRA logging expectations
- Privacy & transparency considerations

9. Use-Cases & Detection Scenarios

Memory-only malware

- Fileless persistence
- Suspicious subnet behavior
- CPU-behavior anomalies
- Supply-chain tampering
- CRA-aligned detection coverage

11. Conclusion

- Summary of alignment
- Future improvements
- Notes for CRA auditors
- Independent author statement

12. Appendices

- A: Glossary
- B: Signal Types
- C: Killing Chain Phase Table
- D: Rule Format Reference
- E: Supply-Chain Module Reference

Appendix A: Glossary

Behavior Signal: Signal derived from observable system behavior (process actions, network flows, CPU patterns, memory usage, user complaints) used for detection without relying on file hashes or static signatures.

- **M-Module (Mathematical Module):** Public mathematical structure in VNAGY that ingests signals, normalizes them, and produces behavior scores using transparent, weight-free logic suitable for offline-first environments.
- **Killing Chain:** VNAGY's internal phase model for describing the lifecycle of an attack (from initial foothold to impact), mapped to MITRE ATT&CK and used for correlation and rule design.
- **Heuristic Scoring:** Scoring method that uses rules, thresholds, and contextual logic (rather than ML weights) to assign risk levels to behaviors in an explainable way.
- **Memory-Only Detection:** Detection of attacks that operate primarily in RAM (no traditional files on disk), focusing on process behavior, allocation patterns, and execution traces.
- **Subnetting Heuristic:** Heuristic that monitors subnet-level behavior (e.g., unusual lateral movement, scanning, or traffic concentration) to detect anomalies in local network segments.
- **CPU-Behavior Monitoring:** Observation of CPU usage patterns (spikes, unusual instruction mix, sustained abnormal load) as a behavior signal for detecting suspicious or malicious activity.

- **Supply-Chain Guard:** VNAGY's set of controls that protect build, deployment, and dependency processes from tampering, including hash checks, signature validation, and offline builds.
- **Offline-First Processing:** Design principle where all critical detection, scoring, and logging functions operate locally without requiring continuous network or cloud connectivity.

Appendix B: Signal types

VNAGY's Behavior Engine uses five primary signal families:

- **Process Signals** — process creation/termination, parent-child relationships, command-line arguments, privilege changes, unusual execution paths.
- **Network Signals** — outbound/inbound connections, burst patterns, subnet anomalies, unusual ports or protocols.
- **CPU-Behavior Signals** — abnormal load, irregular thread scheduling, unusual instruction patterns (where observable).
- **Memory-Only Signals** — execution from non-standard memory regions, reflective loading, suspicious allocation/free patterns, processes without stable disk files.
- **User Complaint Signals** — user-reported anomalies (slowdowns, strange pop-ups, unexpected network behavior), treated as contextual behavior input.

Appendix C — Killing Chain Phase Table

Phase	Description	MITRE Mapping	KEY Tools
Initial Foothold	First entry into the system	Initial Access	Network heuristics, behavior rules
Establishment	Persistence and stabilization	Persistence, Priv. Escalation	Process heuristics, memory signals
Lateral Movement	Movement across hosts or subnets	Lateral Movement	Subnetting heuristics, network scoring
Command & Control	Remote control, beaconing, coordination	Command and Control	Network behavior engine
Impact	Exfiltration, encryption, sabotage, disruption	Exfiltration, Impact	CPU/memory behavior, final scoring

This table is simplified for **CRA** documentation; **VNAGY** may internally use more granular phases.

Appendix D — Rule Format Reference (Ultra-Minimal Version)

Rule Structure:

- Rule ID
- Name
- Phase
- Signals
- Conditions
- Score Impact
- Actions

Example: Suspicious Subnet Scan — network burst + single-process dominance → medium risk → log + alert.

Appendix E — Killing Chain Format (Ultra-Minimal Version)

Killing Chain Structure:

- **Phase ID**
- **Phase Name**
- **Description**
- **MITRE Mapping**
- **Key Behavior Signals**

Phases (Minimal):

- **Foothold** — initial access
- **Establishment** — persistence and privilege gain
- **Lateral Movement** — subnet traversal
- **C2** — remote coordination
- **Impact** — exfiltration or disruption

Usage: Killing Chain phases provide a deterministic reference for rule design, behavior scoring, and **CRA**-aligned documentation.

13. Intellectual Property & Licensing

13.1. Ownership

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13.3. Non-Operational Declaration

This document contains no operational details, no thresholds, no weights, no heuristics, no correlation logic, and no algorithmic semantics. All constructs are symbolic and conceptual. The model is non-computational and non-reconstructable by design.

13.5. License

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